

MORRISVILLE-STOWE, VT, USAWMO: **726114**Lat: **44.534N**Lon: **72.614W**Elev: **732**StdP: **14.31**Time Zone: **-5.00 (NAE)**Period: **97-19**WBAN: **54771****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-6.9	-19.8	1.9	-10.6	-14.8	2.5	-5.2	18.9	19.6	16.9	18.2	1.7	200	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	86.4	70.4	83.5	69.0	81.1	67.5	73.1	81.9	71.4	79.8	69.8	77.3	6.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	114.8	77.1	68.6	108.1	75.7	66.6	100.7	74.0	37.1	82.0	35.6	80.0	34.3	77.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.0	12.1	-21.2	91.3	6.4	2.4	-25.8	93.0	-29.5	94.3	-33.1	95.7	-37.7	97.4		
(5)			-21.6	76.5	6.2	2.0	-26.0	77.9	-29.6	79.1	-33.1	80.2	-37.5	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	17.1	19.9	28.3	41.8	54.4	62.9	67.6	66.0	59.3	46.7	35.4	24.1
	DBStd	19.90	12.95	11.32	11.10	8.52	7.84	6.58	5.25	5.56	7.48	8.23	9.24	11.18
	HDD50	4281	1020	844	676	274	43	0	0	0	10	166	445	803
	HDD65	8057	1485	1264	1138	696	341	117	32	57	203	568	889	1267
	CDD50	1998	0	0	3	28	179	387	547	496	288	63	6	1
	CDD65	299	0	0	0	1	11	53	114	88	31	1	0	0
	CDH74	3073	0	0	5	38	211	612	1099	785	311	12	0	0
	CDH80	749	0	0	0	8	46	158	285	176	76	0	0	0
(14) Wind	WSAvg	4.1	4.3	4.6	5.2	5.1	4.4	3.7	3.2	2.8	3.1	3.9	4.1	4.2
(15) Precipitation	PrecAvg	50.1	3.0	2.8	3.1	3.5	4.0	5.0	5.6	4.5	4.1	4.9	3.8	4.1
	PrecMax	58.2	5.4	5.0	5.1	8.9	7.2	10.3	9.8	10.2	8.5	9.3	5.6	8.3
	PrecMin	39.6	1.5	1.2	1.3	1.0	1.7	1.3	3.1	1.3	1.8	1.9	1.5	1.9
	PrecStd	4.0	1.0	0.9	1.0	1.9	1.7	2.3	1.6	1.9	1.5	2.1	1.2	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	51.8	65.8	79.2	85.6	89.4	90.0	88.7	87.7	75.5	65.2	54.8
		MCWB	46.1	43.5	54.5	60.2	66.0	71.2	73.8	72.5	70.5	62.7	56.9	47.3
	2%	DB	42.5	44.5	54.8	70.7	79.9	84.2	86.2	84.2	81.4	70.3	59.7	48.3
		MCWB	38.8	38.2	45.2	54.9	61.6	68.5	71.2	69.9	68.0	59.6	53.3	44.5
	5%	DB	37.8	39.3	49.3	63.9	75.1	81.2	83.1	81.5	77.3	65.4	54.4	42.9
		MCWB	34.6	35.0	41.5	51.1	59.6	67.0	69.3	68.6	65.6	57.0	48.3	39.2
	10%	DB	35.1	36.3	44.2	57.5	70.8	77.3	80.7	78.9	72.9	61.1	49.3	38.4
		MCWB	32.7	32.9	38.2	47.3	57.8	64.4	68.0	67.1	63.9	54.7	44.2	35.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	46.0	55.5	62.9	69.3	73.8	76.3	74.7	72.7	65.9	58.4	49.6
		MCDB	49.5	49.7	65.7	75.3	79.1	83.6	86.2	83.7	81.7	71.2	63.0	54.1
	2%	WB	39.4	39.2	46.7	57.3	65.9	71.5	73.6	72.2	70.0	61.7	53.7	44.8
		MCDB	41.8	42.9	52.1	67.2	75.4	80.6	81.7	80.9	78.1	67.1	59.2	48.0
	5%	WB	35.4	36.0	42.2	53.0	62.9	69.2	71.6	70.5	67.6	58.3	49.2	39.7
		MCDB	37.3	39.1	48.6	61.1	70.4	77.3	79.7	78.1	75.2	64.4	53.5	42.7
	10%	WB	32.7	33.4	38.7	48.4	60.1	66.9	69.8	68.8	65.3	54.9	45.1	35.7
		MCDB	34.9	35.8	43.5	57.1	67.7	74.0	77.3	75.6	71.7	59.8	48.8	38.1
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	20.5	19.7	21.3	23.1	21.8	21.8	22.0	22.4	18.8	16.0	15.5
		MCDBR	18.8	22.6	26.4	34.3	32.3	27.9	25.4	24.8	27.2	26.3	24.5	17.4
		MCWBR	16.5	17.9	17.8	19.5	16.8	13.9	12.2	12.3	14.5	16.3	18.2	14.7
	5% WB	MCDBR	16.7	20.7	23.9	29.6	24.9	23.2	21.8	21.6	22.8	22.6	22.1	16.8
		MCWBR	15.2	17.3	17.3	18.4	14.5	12.5	11.4	11.4	12.9	15.4	17.8	14.8
(40) Clear-Sky Solar Irradiance	taub		0.270	0.284	0.323	0.351	0.372	0.424	0.436	0.408	0.365	0.330	0.306	0.277
	taud		2.375	2.315	2.341	2.387	2.389	2.240	2.215	2.284	2.401	2.504	2.497	2.426
	Ebn at Noon		272	288	289	289	285	269	264	267	272	269	257	258
	Edh at Noon		24	30	34	36	37	43	44	39	32	25	22	21
(44) All-Sky Solar Radiation	RadAvg		443	715	1066	1337	1571	1712	1745	1574	1263	737	458	324
	RadStd		56	77	70	165	172	154	132	78	96	80	61	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	N/A

Nomenclature: See separate page